

Course Code	STA8 EJ 412				
Course Title	OPERATIONS RESEARCH				
Type of Course	Major Elective				
Semester	VIII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Good idea about Linear Programming Problems				
Course Summary	Understand advanced models of Linear Programming Problems and Non-Linear Programming Problems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
1.	Express theoretical knowledge of Simplex method for solving Linear Programming Problems and Expertise with Revised Simplex method and Dual Simplex method.	U	F	Seminar Presentation /
2.	Develop and solve Integer Programming Models.	Ap	C	Instructor-created exams
3.	Identify Classical Optimization techniques and Non- linear Programming techniques.	U	F	Group Tutorial Work
4.	Formulate Network models like PERT and CPM to improve decision making and develop critical thinking objective analysis of decision making.	Ap	C	Home Assignments
5.	Develop and solve Inventory Models and acquire skills in analyzing Queuing Models.	Ap	C	Practical Skill
6.	Estimate the allocation of resources to demand points in the best possible way using various techniques and minimize the cost or time of completion of jobs.	An	P	Observation of Practical Skills
* - Remember (R), Understand (U), Apply (Ap), Analyze (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive				

Detailed Syllabus:

Module	Unit	Content	Hrs (60)	Marks (70)
I	Theory of Simplex Method & Revised Simplex Method		12	20
	1	Canonical and Standard form of LP problem		
	2	Canonical and Standard form of LP problem- Reduction of Feasible solution to Basic Feasible solution		
	3	Improving a Basic Feasible Solution-Alternative optimal solutions		
	4	Unbounded Solutions-Unrestricted variables–degeneracy and its Resolution		
	5	Standard forms for Revised Simplex Method- Computational Procedure		
	6	Comparison of Simplex method and Revised Simplex method		
	7	Dual Simplex Method		
II	Integer Linear Programming & Classical Optimization Methods		12	20
	8	Types of Integer Programming Problems-Gomory's all Cutting Plane Method		
	9	Gomory's Mixed Integer Cutting Plane Method		
	10	Branch and Bound Method		
	11	Applications of Zero-One Integer Programming		
	12	Unconstrained Optimization- Optimizing single variable and Multivariable functions		
	13	Constrained Multi Variable Optimization with equality and Inequality constraints		
	14	Lagrange Multipliers Methods		
III	Non-Linear Programming Methods, Quadratic Programming & Dynamic Programming		12	15
	16	The General Non-Linear Programming Problem- Graphical Solution Method		
	17	Quadratic Programming -Kuhn-Tucker Conditions- Wolfe's Modified Simplex Method		
	18	Dynamic Programming -Terminology -Optimal Decision Policy-General Algorithm-		
	19	Dynamic Programming Approach for solving LPP		
IV	Project Management PERT and CPM, Inventory Control Models		12	15
	20	Basic difference between PERT and CPM-Critical Path Analysis		
	21	Estimation of Project completion time- Project Time cost Trade off -Project Crashing -Resource allocation		
	22	Deterministic Inventory Models- EOQ Inventory Models without shortages and with Shortages-		
	23	Probabilistic Models-Newspaper Boy Problem.		
V	Open –Ended		12	

	Sequencing Problem, Replacement and Maintenance Models Simulation Techniques		
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Reference

1. **Mital. K. V. and Mohan. C. (1996).** Optimization Methods in Operations Research and Systems Analysis Third Edition, New Age International (Pvt) Ltd., New Delhi.
2. **Taha. H.A. (2007).** Operations Research – An Introduction-Eighth Edn. Pearson Printice Hall, new Jersey.
3. **Sharma J.K. (2003).** Operations Research-Theory and Applications, Macmillan Indian Ltd., New Delhi
4. **Man Mohan, Kanti Swarup and Gupta (1999).** Operations Research, Sulthan Chand & Sons, New Delhi.